

Design of Direct Digital Frequency Synthesizer With High ROM Compression Ratio

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Abstract—A new ROM size reduction technique for direct digital frequency synthesizers (DDFS's) is proposed in this paper. In the new technique, a multiple-segment piecewise polynomial function $g(x)$ is determined for the sinusoidal ROM table which stores the values of $\sin(\pi x/2) - g(x)$ instead of $\sin(\pi x/2)$. With use of the proposed three-segment technique for 20-bit phase to 11-bit amplitude mapping, the number of ROM output bits and ROM size are reduced by 5 and 98.58%, respectively, compared to the quarter sine wave technique with comparable spectral purity. The compression ratio of the proposed technique is about 70 : 1. This new design can gain low power and small chip area, especially for high frequency resolution applications.

I. INTRODUCTION

Fast frequency switching with fine frequency steps is crucial in modern wireless communications. The traditional phase-locked loop (PLL) based synthesizer is not suitable in wireless communications due to the inherent loop delay. Direct digital frequency synthesizers (DDFSs) [1-7] play a central role in modern high-performance communication systems because of their unique characteristics like fast switching speed, fine frequency resolution, low phase noise, and high spectral purity. It consists of a phase accumulator and a sine/cosine generator. The phase accumulator acts as a slope-controlled digital sweep generator. The core of a DDFS is the sine/cosine generator block.

Many applications simply use a ROM lookup table method to implement the generator block. This method may require a large ROM to store the sinusoidal samples. Many techniques to reduce the size of the ROM lookup table have been proposed such as the Sunderland algorithm [1], the CORDIC algorithm [2], ROM compression methods [3, 4], and so on. The Sunderland algorithm was based on simple trigonometric approximation to reduce the ROM size. The CORDIC algorithm, which uses an iterative computation method, can reduce the required ROM size efficiently, but the computational complexity is also large. The ROM compression method proposed by Chen et al. [3] chose the function $1.375x$ and $1 - 0.375x + h(x)$ to approximate

$\sin(\pi x/2)$ and $\cos(\pi x/2)$ within $[0, \pi/4]$, where $h(x)$ is defined as $0.25x$ when $x \leq 0.25$ and $0.125 - 0.25x$ when $x > 0.25$. But it is only useful for the range $[0, \pi/4]$, however, the required range for quarter wave symmetry is $[0, \pi/2]$.

In this paper, the proposed technique incorporates some ROM compression methods, such as quarter wave symmetry, the Sunderland algorithm, and sine-phase difference algorithm to obtain better compression ratio. The ROM table in the new technique is modified by $\sin(\pi x/2) - g(x)$ instead of $\sin(\pi x/2)$. The function $g(x)$ can be segmented into multiple pieces in the range $[0, \pi/2]$. We determined the optimal segment intervals by minimizing the total area or minimizing the maximum value between the sine function and $g(x)$. The new technique reduces the ROM size obviously. The proposed piecewise function $g(x)$ can be implemented using only adders and multiplexers without multipliers.

II. DDFS AND ROM COMPRESSION METHODS

In DDFS systems, the common DDFS architecture consists of an overflowing accumulator which is increased by a number, called frequency control word (FCW), for each clock cycle. The accumulated result is used as the address for a sine/cosine generator, generally a lookup table in where the sine/cosine sample values are stored. And a ROM-based lookup table is used to compute the sine/cosine function or another function you want. Since the fix-length accumulator regularly overflows, the phase is periodic. For a given accumulator word length L and FCW, the relation between the generated wave frequency and the clock frequency f_{clk} is [6]:

$$f_{out} = FCW \cdot \frac{f_{clk}}{2^L}. \quad (1)$$

The frequency resolution of the DDFS is:

$$\Delta f = \frac{f_{clk}}{2^L}. \quad (2)$$

The ROM size grows exponentially as the frequency resolution increases. Therefore, an important design issue for DDFS is the ROM compression technique. Some methods are described as follows.

A. Quarter Wave Symmetry

For digital frequency synthesizers, a well known technique is to store only $\pi/2$ rad of sine/cosine information because of the symmetric property of the sine/cosine function, and to generate the sine lookup table samples for the full range of 2π by exploiting the quarter wave symmetry of the sine function. The reduction of lookup table capacity increases the additional logic circuits required to generate the complement of the accumulator and lookup table output, as shown in Figure 1. The fundamental method to generate sine waveform in a 2π period is to collect the quadrant of sine function between 0 and $\pi/2$, and synthesize sine function between 0 and π by taking the symmetry of the quadrant. The full sine wave can be obtained from the sine function between 0 and π by sign inversion. Hence, the compression of sine function samples between 0 and $\pi/2$ is the first important issue in the DDFS.

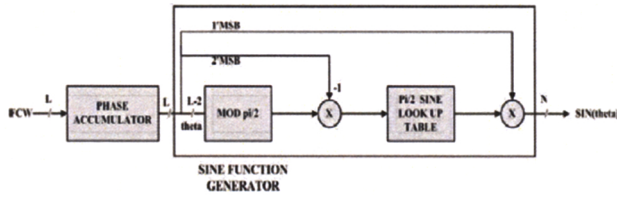


Figure 1. Block diagram of the quarter wave symmetry method

B. Sunderland Algorithm

The Sunderland architecture is a well-known technique to reduce the size of the ROM table. It comprises a coarse ROM, a fine ROM, and an adder. Incorporating the quarter wave symmetry mentioned in section A, the two most significant bits are used to decode the quadrant, while the remaining bits are used to address a one-quadrant ROM. The most significant bit determines the required sign of the result, and the next most significant bit controls whether the phase between 0 and $\pi/2$ should be increased or decreased. The phase accumulator value θ is decomposed into three fractions as $\theta = \alpha + \beta + \gamma$ where γ represents the least signification bits, β the middle bits, and α the most signification bits, and $\alpha < \pi/2$, $\beta < (\pi/2) \cdot 2^{-A}$, $\gamma < (\pi/2) \cdot 2^{-(A+B)}$ with the word lengths of variables α , β , and γ are A, B, and C, respectively. The architecture provides a reduction in the lookup table storage by replacing the storage requirements of one large ROM of size 2^{A+B+C} words with two smaller ROMs of size 2^{A+B} words and 2^{A+C} words, whose outputs are added together to construct the sine

function. By the trigonometric identities, the desired sine function may be approximated by

$$\begin{aligned} \sin\left(\frac{\pi}{2}(\alpha + \beta + \gamma)\right) &= \sin\left(\frac{\pi}{2}(\alpha + \beta)\right)\cos\left(\frac{\pi}{2}\gamma\right) + \cos\left(\frac{\pi}{2}(\alpha + \beta)\right)\sin\left(\frac{\pi}{2}\gamma\right) \\ &\approx \sin\left(\frac{\pi}{2}(\alpha + \beta)\right) + \cos\left(\frac{\pi}{2}(\alpha + \beta)\right)\sin\left(\frac{\pi}{2}\gamma\right). \end{aligned} \quad (3)$$

The equation is implemented by a coarse ROM table and a fine ROM table by storing the value $\sin[\pi(\alpha + \beta)/2]$ and $\cos(\pi\alpha/2)\sin(\pi\gamma/2)$, respectively.

C. Sine-Phase Difference (SPD) Algorithm

The sine-phase difference algorithm has been published after the Sunderland's paper. The compression technique has been used where $\sin(\pi\theta/2) - \theta$ instead of the quarter-wave sine function is stored in the ROM. The penalty for this storage reduction is the introduction of an extra adder at the output of the lookup table to perform the following operation:

$$f(\theta) = \left\{ \sin\left(\frac{\pi}{2}\theta\right) - \theta \right\}_{ROM} + \theta \quad (4)$$

where the subscript *ROM* denotes that the values in the brace are stored in the ROM lookup table. And we have

$$\max \left[\sin\left(\frac{\pi}{2}\theta\right) - \theta \right] \approx 0.21 \max \left[\sin\left(\frac{\pi}{2}\theta\right) \right]. \quad (5)$$

From (5), two bits of amplitude (about 1/4) in the storage for the sine function are saved.

III. THE PROPOSED THREE-SEGMENT PIECEWISE FUNCTION METHOD

Incorporating the three methods mentioned in section II, a new method to compress the ROM table of the sine/cosine wave called "piecewise function method" is proposed to obtain the best performance. As shown in Figure 2, consider the piecewise function $g(x)$ composed of three segments in the range $[0, \pi/2]$ to define the new storage requirement of $\sin(\pi x/2) - g(x)$ instead of $\sin(\pi x/2) - x$ in order to reduce the ROM size.

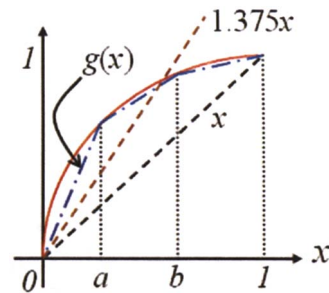


Figure 2. The proposed three-segment piecewise function algorithm

The optimal values, a and b , are determined by minimizing the total area or minimizing the maximum value between the sine function and $g(x)$. The slopes in the three intervals are shown in Figure 3. For the piecewise function $g(x)$ found by choosing the optimal piecewise interval points, a and b , that minimizes the total area in the range $[0, \pi/2]$, we have

$$[a, b] = \arg \min_{a, b} \sum_{x=0}^a \left[\sin\left(\frac{\pi}{2} \cdot x\right) - \frac{1}{a} \sin\left(\frac{\pi}{2} \cdot a\right) x \right] + \sum_{x=a}^b \left[\sin\left(\frac{\pi}{2} \cdot x\right) - \left[\frac{1}{b-a} \left(\sin\left(\frac{\pi}{2} \cdot b\right) - \sin\left(\frac{\pi}{2} \cdot a\right) \right) (x-a) + \sin\left(\frac{\pi}{2} \cdot a\right) \right] \right] + \sum_{x=b}^1 \left[\sin\left(\frac{\pi}{2} \cdot x\right) - \left[\frac{1}{1-b} \left(1 - \sin\left(\frac{\pi}{2} \cdot b\right) \right) (x-b) + \sin\left(\frac{\pi}{2} \cdot b\right) \right] \right]. \quad (6)$$

The values, a and b , are obtained as 0.407 and 0.715, respectively. Another way to choose the optimal interval points by minimizing the maximum value between the sine function and $g(x)$ is

$$[a, b] = \arg \min_{a, b} \left[\max \left[\sin\left(\frac{\pi}{2} \cdot x\right) - g(x) \right] \right]. \quad (7)$$

The values, a and b , are obtained as 0.43 and 0.73, respectively. The values of a and b found by the above two criteria have similar performance in our design.

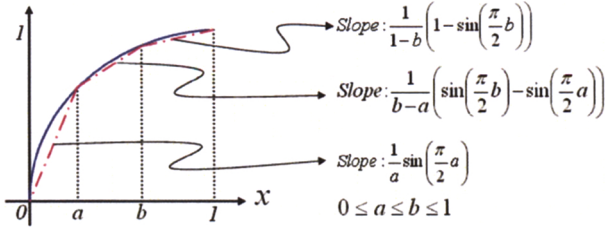


Figure 3. The slopes for the three segments of the proposed method

The multiple-segment piecewise function method can significantly reduce the ROM size compared to the sine-phase difference algorithm. Furthermore, incorporating the Sunderland architecture and the quarter wave symmetry method can make the compression ratio more efficiently. Using the sine-phase difference algorithm to modify the Sunderland architecture, the new architecture is shown in Figure 4. The values stored in coarse ROM and fine ROM are $\sin[\pi(\alpha + \beta)/2] - (\alpha + \beta)$ and $\cos(\pi\alpha/2)\sin(\pi\gamma/2)$, respectively. Comparing the ideal sine wave to the above three methods (Sunderland algorithm, sine-phase difference algorithm, and piecewise function algorithm), the required storages for ROM are shown in Figure 5. It is obvious that the total area stored in ROM using the piecewise function method is smaller than the other methods.

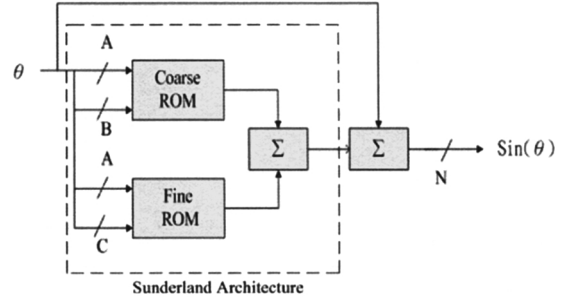


Figure 4. The modified Sunderland architecture with the sine-phase difference algorithm

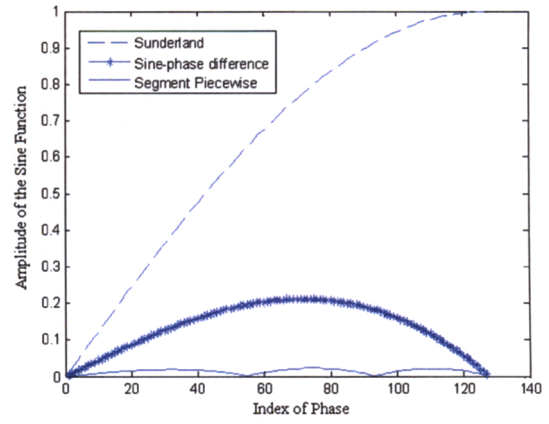


Figure 5. Comparison of the required ROM storage

IV. IMPLEMENTATION AND VERIFICATION

The ROM architecture of the three-segment piecewise function method that we proposed is depicted in Figure 6. It includes the following blocks:

- A multiplexer selecting a segment through input phase.
- Three different shift and add operations.
- A coarse ROM and a fine ROM by use of the Sunderland architecture.

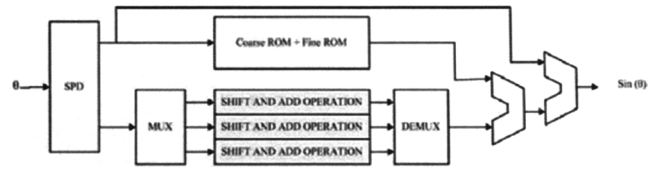


Figure 6. Architecture of the three-segment piecewise function method

The above three methods (Sunderland algorithm, sine-phase difference algorithm, and the three-segment piecewise function algorithm) are implemented for the same case where the accumulator word length $L=20$, $FCW=11$ bits, the lengths of the three phase fractions $A=B=C=4$, and the

output word length $N=11$ bits. The comparison of ROM size and compression ratio is listed in TABLE I. The mean square error (MSE) and spurious loss for each method are designed to be about $10E-3$ and 75 dBc for the comparison purpose. The result shows that the three-segment piecewise function method achieves higher compress ratio.

Some equipment such as pattern generator (PG), logic analyzer (LA), and Xilinx Virtex-II FPGA are adopted for hardware implementation and verification. The design flow is shown as Figure 7.

TABLE I. Comparison of ROM Size and Compression Ratio

Method	Non-compressed	Sunderland	Sine-Phase Difference	Segment Piecewise
CoarseROM output		11 bits	9 bits	6 bits
FineROM output		5 bits	5 bits	4 bits
ROM Size	180224	4096	3584	2560
ROM Size percentage	100%	2.27%	2%	1.42%
compression ratio	1:1	44:1	50:1	70:1

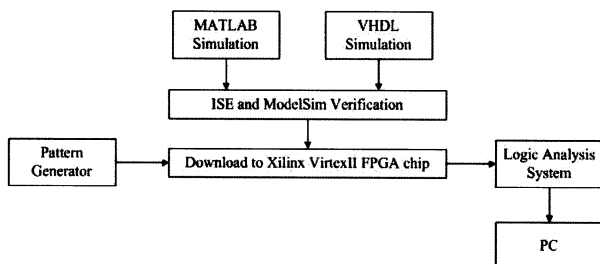


Figure 7. The flow chart of implementation and verification

We use Xilinx Virtex-II XC2V1000 FPGA to verify our design. The FCW is controlled by PG. Then LA is used to grab the output data for analysis. A DDFS output example snapped from the LA is shown in Figure 8. The output data of the designed DDFS is then analyzed by the MATLAB tool to calculate the spectrum. The result is shown in Figure 9, where the spurious loss is about 75.2 dBc as we design. The occupied FPGA resources by the ISE XST synthesizer are listed in TABLE II.

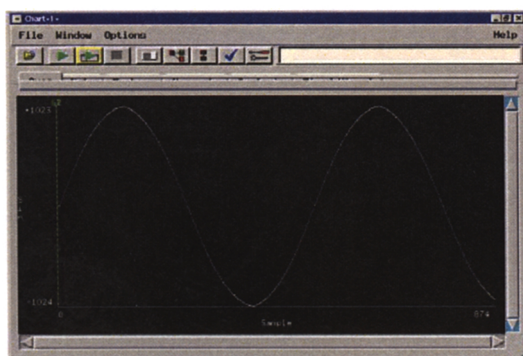


Figure 8. Implementation result snapped from LA

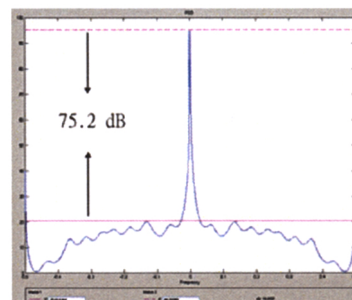


Figure 9. Spurious loss for DDFS output spectrum

TABLE II. FPGA Resources (ISE XST Synthesizer)

Method	Sunderland	Sine-Phase Difference	Segment Piecewise
Number used as 16x1 ROMs	208	208	176
Total equivalent gate count	8218	8236	7993
Number used of slices	144	146	176

V. CONCLUSIONS

The multiple-segment piecewise function method is proposed as an efficient ROM compression method for the DDFS design. The new DDFS has been modeled with VHDL and verified on Xilinx FPGA. With use of the proposed three-segment technique for 20-bit phase to 11-bit amplitude mapping, the number of ROM output bits and ROM size are reduced by 5 and 98.58%, respectively, compared to the quarter sine wave technique with comparable spectral purity. The compression ratio of the proposed technique is 70 : 1. This proves that the proposed technique exhibits an attractive potential to decrease the ROM size.

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